



Summer Internship Completion Report

on

Study of Gene Expression Research in Asthma

By

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At

**DR. D. Y. PATIL BIOTECHNOLOGY AND BIOINFORMATICS
INSTITUTE, DR. D. Y. PATIL VIDYAPEETH, PUNE**

(Deemed to be University)

(Accredited (3 rd cycle) by NAAC with a CGPA of 3.64 on four-point scale at 'A++' Grade)

(Declared as Category - I University by UGC under Graded Autonomy Regulations, 2018)

(An ISO 9001: 2015 and 14001:2015 Certified University and Green Education Campus)

Approved by AICTE, Govt. of India; DST-FIST Supported Institute

Approved as Acharya Chanakya Kaushal Vikas Kendra by Ministry of Skill Development and Entrepreneurship, Govt. of India

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ACKNOWLEDGEMENT:

I would like to express my sincere gratitude to **Dr. D. Y. Patil Biotechnology and Bioinformatics Institute, Dr. D. Y. Patil Vidyapeeth, Pune**, for providing me with the opportunity to undergo this Bioinformatics Summer Internship. The internship offered an excellent platform to understand the integration of biology with computational tools and significantly strengthened my practical knowledge of modern bioinformatics.

I extend my heartfelt thanks to the Director, Institutional Coordinators, faculty members, and mentors for designing a comprehensive training program that combined theoretical concepts with hands-on computational analysis. Their continuous guidance, encouragement, and valuable insights helped me understand the practical applications of bioinformatics in biological and biomedical research.

I am especially grateful to my mentors for patiently explaining complex concepts related to transcriptomics, structural bioinformatics, molecular docking, biological databases, and computational analysis. Their support motivated me to explore scientific problems with curiosity and confidence.

I would also like to thank my home institute, **Motilal Nehru National Institute of Technology (MNNIT), Prayagraj**, for granting me the opportunity to participate in this internship and encouraging experiential learning beyond the classroom.

Finally, I sincerely thank my parents, friends, and everyone who supported and motivated me throughout the internship. Their encouragement played an important role in helping me successfully complete this training and prepare this report.

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TITLE:

Integrated Bioinformatics Approaches for Gene Expression Analysis, Functional Annotation, Protein Structure Prediction and Molecular Docking: A Summer Internship Report

INTRODUCTION:

Bioinformatics is an interdisciplinary field that combines biology, computer science, statistics, mathematics, and information technology to analyze, interpret, and manage biological data. With the rapid advancement of high-throughput technologies such as next-generation sequencing (NGS), microarrays, and high-resolution structural biology techniques, biological research has shifted from studying individual genes to understanding entire biological systems through computational approaches.

Modern biomedical research generates enormous amounts of genomic, transcriptomic, proteomic, and structural data. Bioinformatics provides the computational tools and databases necessary to organize, analyze, and interpret this information efficiently. These analyses help researchers understand disease mechanisms, identify biomarkers, discover therapeutic targets, predict protein structures, and accelerate drug discovery.

During this internship, I gained practical exposure to a complete bioinformatics workflow beginning with biological databases and gene expression analysis, followed by functional enrichment, protein interaction network analysis, structural bioinformatics, homology modelling, molecular visualization, molecular docking, and virtual screening. Rather than studying each tool independently, the internship demonstrated how these computational resources are connected to solve real biological problems.

A major component of the internship involved the analysis of publicly available asthma transcriptomic datasets from the NCBI Gene Expression Omnibus (GEO). Differentially expressed genes were identified using GEO2R and subsequently analyzed using STRING, Cytoscape, Gene Ontology (GO), KEGG Pathway Database, UniProt, Protein Data Bank (PDB), AlphaFold Protein Structure Database, SWISS-MODEL, and PyMOL to understand their biological significance and structural characteristics.

In addition to computational biology, the internship also included workshops on scientific writing, research methodology, biostatistics, laboratory safety, medicinal chemistry, stereochemistry, and molecular docking. These sessions provided a broader understanding of how computational methods integrate with experimental biology and pharmaceutical research. Overall, this internship served as an excellent opportunity to strengthen my understanding of bioinformatics workflows while developing practical skills in biological data analysis, structural biology, computational drug discovery, scientific communication, and research interpretation.

AIM:

The primary aim of this internship was to develop a comprehensive understanding of bioinformatics by integrating biological knowledge with computational tools used in modern biomedical research. The internship aimed to provide hands-on training in biological database exploration, gene expression analysis, functional annotation, protein structure prediction, molecular visualization, protein–protein interaction analysis, molecular docking, virtual screening, and scientific data interpretation. Furthermore, it sought to build practical skills in analyzing publicly available biological datasets and understanding their applications in disease research, biomarker discovery, and computational drug discovery while enhancing scientific writing, data presentation, and analytical thinking.

OBJECTIVES:

The major objectives of this internship were:

1. To understand the fundamental concepts of bioinformatics and its applications in biological and biomedical research.
2. To gain practical experience in exploring biological databases such as NCBI, GEO, UniProt, Protein Data Bank (PDB), PDBsum, AlphaFold Protein Structure Database, SWISS-MODEL, Gene Ontology (GO), KEGG Pathway Database, and STRING.
3. To learn the principles of transcriptomics, RNA sequencing, microarray technology, differential gene expression analysis, and their applications in understanding disease mechanisms.
4. To perform gene expression analysis using GEO datasets and GEO2R for identifying differentially expressed genes associated with human diseases.
5. To interpret biological significance through functional enrichment analysis using Gene Ontology (GO), KEGG pathways, and g:Profiler.
6. To construct and analyze Protein–Protein Interaction (PPI) networks using STRING and visualize biological interaction networks using Cytoscape.

7. To understand protein annotation, protein structure determination, homology modelling, and computational prediction of protein structures using UniProt, PDB, SWISS-MODEL, and AlphaFold.
8. To develop practical skills in protein structure visualization and structural analysis using PyMOL.
9. To understand the workflow of molecular docking, virtual screening, protein preparation, ligand preparation, and docking result interpretation using computational drug discovery software.
10. To improve scientific writing, data interpretation, presentation skills, portfolio preparation, and critical thinking required for modern biological research.
11. To integrate biological knowledge with computational approaches for solving real-world research problems and developing a strong foundation for future studies in bioinformatics and computational biology.

METHODOLOGY:

The internship followed a structured learning approach that combined theoretical lectures, practical demonstrations, computational exercises, research-based assignments, and scientific discussions. The methodology was designed to provide a step-by-step understanding of modern bioinformatics workflows, beginning with biological databases and progressing towards structural bioinformatics and computational drug discovery.

Initially, an orientation session introduced the internship objectives, laboratory guidelines, computational resources, evaluation procedures, portfolio preparation, and research expectations. Students were familiarized with the overall workflow that would be followed throughout the internship.

A laboratory safety and fire drill session was conducted to educate students about emergency response procedures, laboratory safety regulations, safe handling of research equipment, evacuation protocols, and appropriate actions during emergency situations. Practical demonstrations of fire safety procedures and emergency preparedness were performed to ensure awareness of laboratory safety practices.

To improve scientific thinking and organization of research work, sessions on mind mapping and portfolio management were conducted. Students learned techniques for organizing scientific concepts, planning research activities, documenting observations systematically, maintaining laboratory records, and preparing professional scientific reports.

The internship then introduced the fundamentals of bioinformatics and biological databases. Various publicly available databases, including NCBI, GEO, UniProt, Protein Data Bank (PDB), PDBsum, AlphaFold Protein Structure Database, SWISS-MODEL, STRING, Gene Ontology (GO), KEGG Pathway Database, and g:Profiler, were explored to understand their organization, biological significance, and applications in research.

A major component of the internship focused on transcriptomics and gene expression analysis. Publicly available asthma microarray datasets were retrieved from the NCBI Gene Expression

Omnibus (GEO). Using GEO2R, differential gene expression analysis was performed to compare healthy and disease samples. Statistical filtering based on adjusted p-values and log₂ fold change was applied to identify significantly differentially expressed genes.

The shortlisted genes were subjected to functional enrichment analysis using Gene Ontology (GO), KEGG pathways, and g:Profiler to identify enriched biological processes, molecular functions, cellular components, and signaling pathways associated with asthma. Protein–Protein Interaction (PPI) networks were subsequently constructed using the STRING database, and interaction networks were visualized using Cytoscape for identifying hub genes and understanding network topology.

Several research papers related to asthma transcriptomics were critically reviewed and presented to understand current research methodologies, gene expression signatures, hub gene identification, pathway analysis, and precision medicine approaches in respiratory diseases.

The internship also included sessions on scientific writing, where principles of research communication, manuscript preparation, scientific ethics, referencing, plagiarism awareness, literature review, and publication standards were discussed to improve academic writing skills.

Fundamental concepts of biology and medicinal chemistry were reinforced through lectures covering biomolecular interactions, stereochemistry, structural isomerism, conformational analysis, molecular geometry, and their relevance in biological systems and pharmaceutical sciences.

Structural bioinformatics formed another major component of the internship. Protein sequences were explored using UniProt, experimentally determined structures were retrieved from the Protein Data Bank (PDB), structural summaries were examined using PDBsum, and predicted protein structures were studied using the AlphaFold Protein Structure Database and SWISS-MODEL. Protein visualization and structural analysis were performed using PyMOL to understand protein architecture, domains, ligand binding regions, secondary structures, and three-dimensional molecular organization.

The internship further introduced computational drug discovery through molecular docking and virtual screening. Protein structures were prepared by removing unwanted molecules, correcting missing residues and coordinates where necessary, optimizing protein structures, and preparing ligands for docking studies. Molecular docking software, including AutoDock Tools, AutoDock Vina, and ChimeraX, was used to understand protein–ligand interactions, binding affinity calculations, docking poses, and virtual screening workflows.

A dedicated workshop on biostatistics introduced the importance of statistical analysis in biological research. Fundamental statistical concepts, hypothesis testing, probability distributions, data interpretation, and graphical representation of scientific data were discussed to strengthen quantitative analysis skills.

Finally, students compiled all practical work, computational analyses, assignments, presentations, and workshop activities into a structured internship portfolio. The report preparation process emphasized scientific documentation, accurate interpretation of results, logical presentation of findings, and effective communication of research outcomes.

PIC 1) workflow diagram only (Complete bioinformatics pipeline):

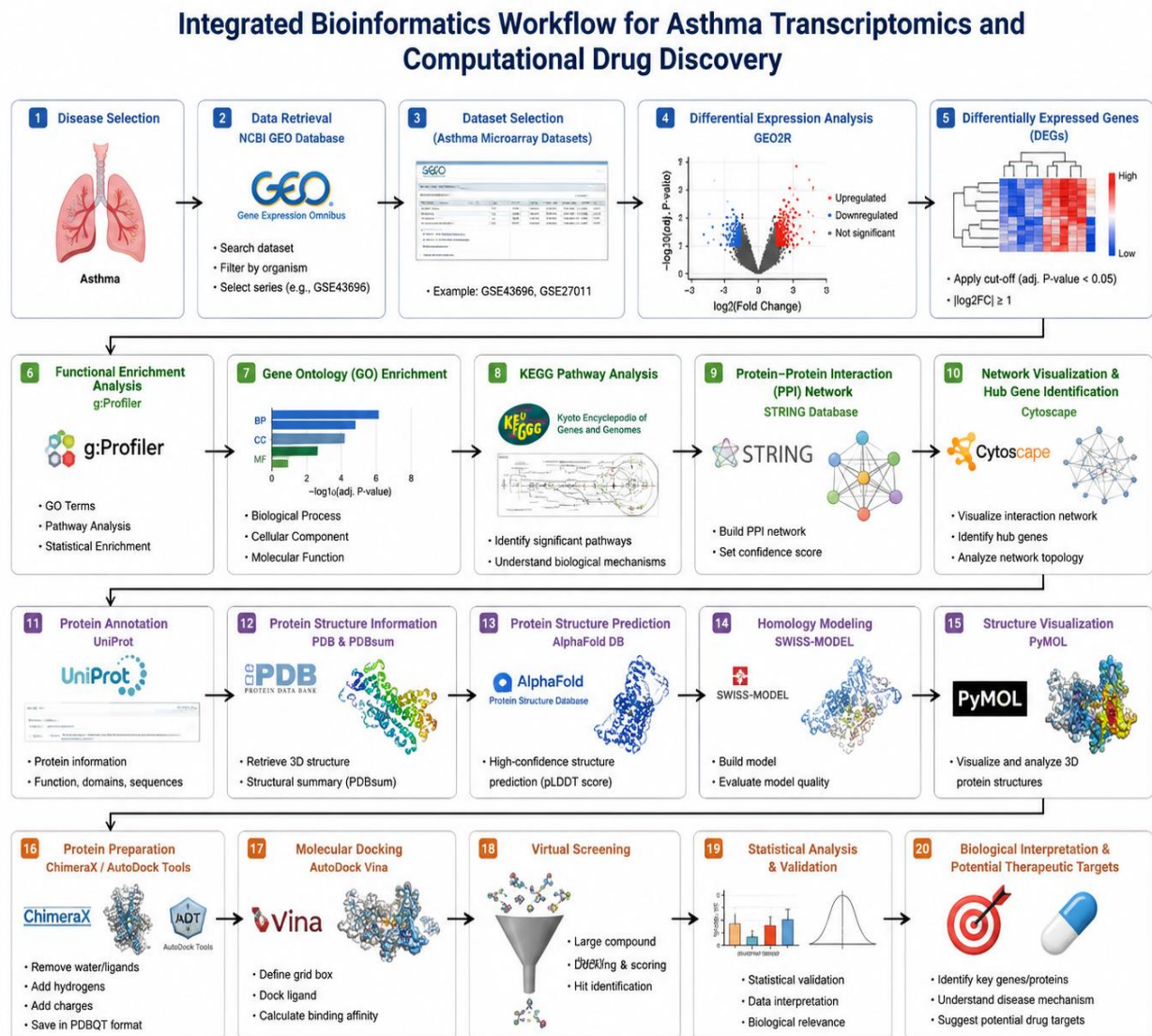


Figure 1. Overall bioinformatics workflow followed during the internship.

ANALYSIS AND INTERPRETATIONS:

The internship demonstrated how bioinformatics integrates multiple computational resources to transform raw biological data into meaningful biological knowledge. Rather than relying on a single database or software, different computational platforms complemented one another to provide functional, structural, and systems-level understanding of biological processes.

Gene expression analysis using GEO datasets enabled the identification of differentially expressed genes associated with disease conditions. Statistical filtering ensured that only biologically relevant genes were selected for downstream analyses, highlighting the importance of appropriate data preprocessing in minimizing false-positive results.

Functional enrichment analysis using Gene Ontology, KEGG, and g:Profiler revealed that biological interpretation extends beyond identifying individual genes. These analyses helped associate differentially expressed genes with biological pathways, cellular functions, immune responses, inflammatory processes, and molecular mechanisms underlying disease progression.

Protein–Protein Interaction analysis using STRING demonstrated that genes rarely function independently. Instead, proteins interact as coordinated biological networks. Cytoscape visualization further improved the understanding of these interactions by enabling identification of hub genes and important network clusters involved in disease development.

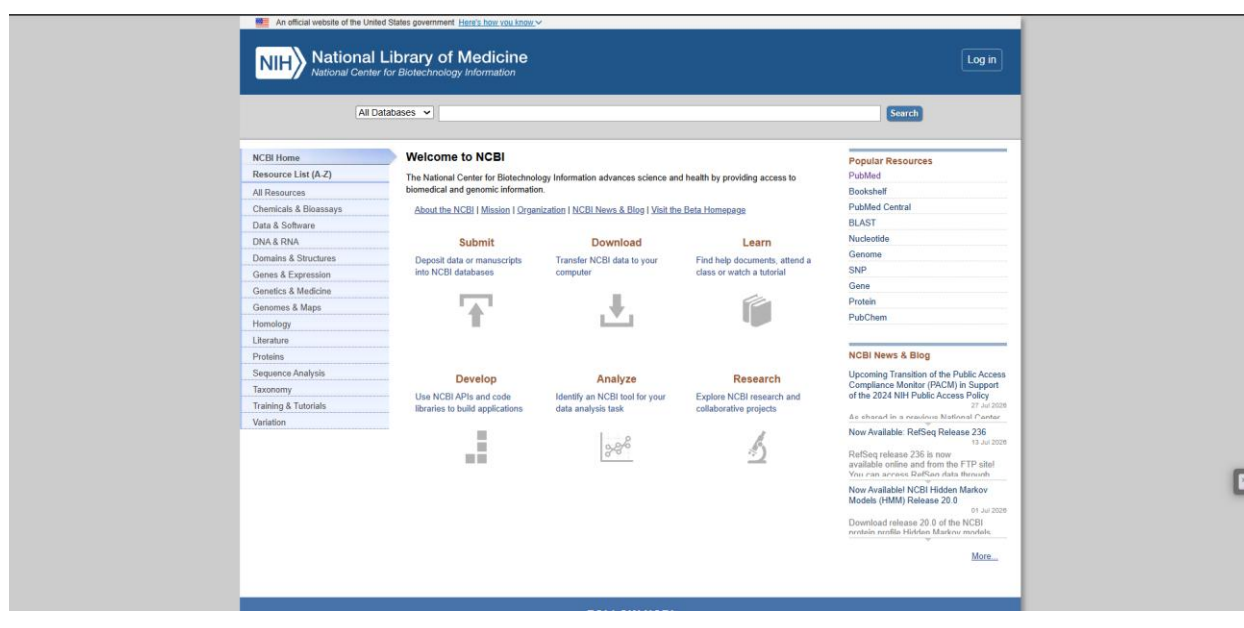
Structural bioinformatics sessions highlighted the importance of protein structure in determining biological function. Exploration of UniProt, PDB, AlphaFold, SWISS-MODEL, and PyMOL illustrated how sequence information, structural prediction, and three-dimensional visualization contribute to understanding protein stability, ligand binding, enzyme activity, and molecular interactions.

The molecular docking and virtual screening modules provided insight into the computational approaches used during early-stage drug discovery. Understanding protein preparation, ligand optimization, docking calculations, binding affinity prediction, and interaction analysis demonstrated how computational techniques significantly reduce experimental workload by prioritizing promising drug candidates before laboratory validation.

The internship also emphasized that computational predictions require careful biological interpretation. Parameters such as statistical thresholds, confidence scores, database evidence, and model quality directly influence research conclusions. This highlighted the importance of critical thinking and scientific validation rather than relying solely on software-generated outputs.

Overall, the internship successfully connected transcriptomics, systems biology, structural bioinformatics, and computational drug discovery into one integrated workflow, demonstrating how bioinformatics supports modern biomedical research from gene discovery to therapeutic target identification.

PIC 2) NCBI Homepage:



PIC 3) NCBI GEO DATASET – GSE43696:


Gene Expression Omnibus

HOME | SEARCH | SITE MAP | GEO Publications | FAQ | MIAME | Email GEO

NCBI > GEO > **Accession Display** [?](#) Not logged in | [Login](#) [?](#)

GEO help: Mouse over screen elements for information.

Scope: Format: Amount: GEO accession:

Series **GSE43696** [Query DataSets for GSE43696](#)

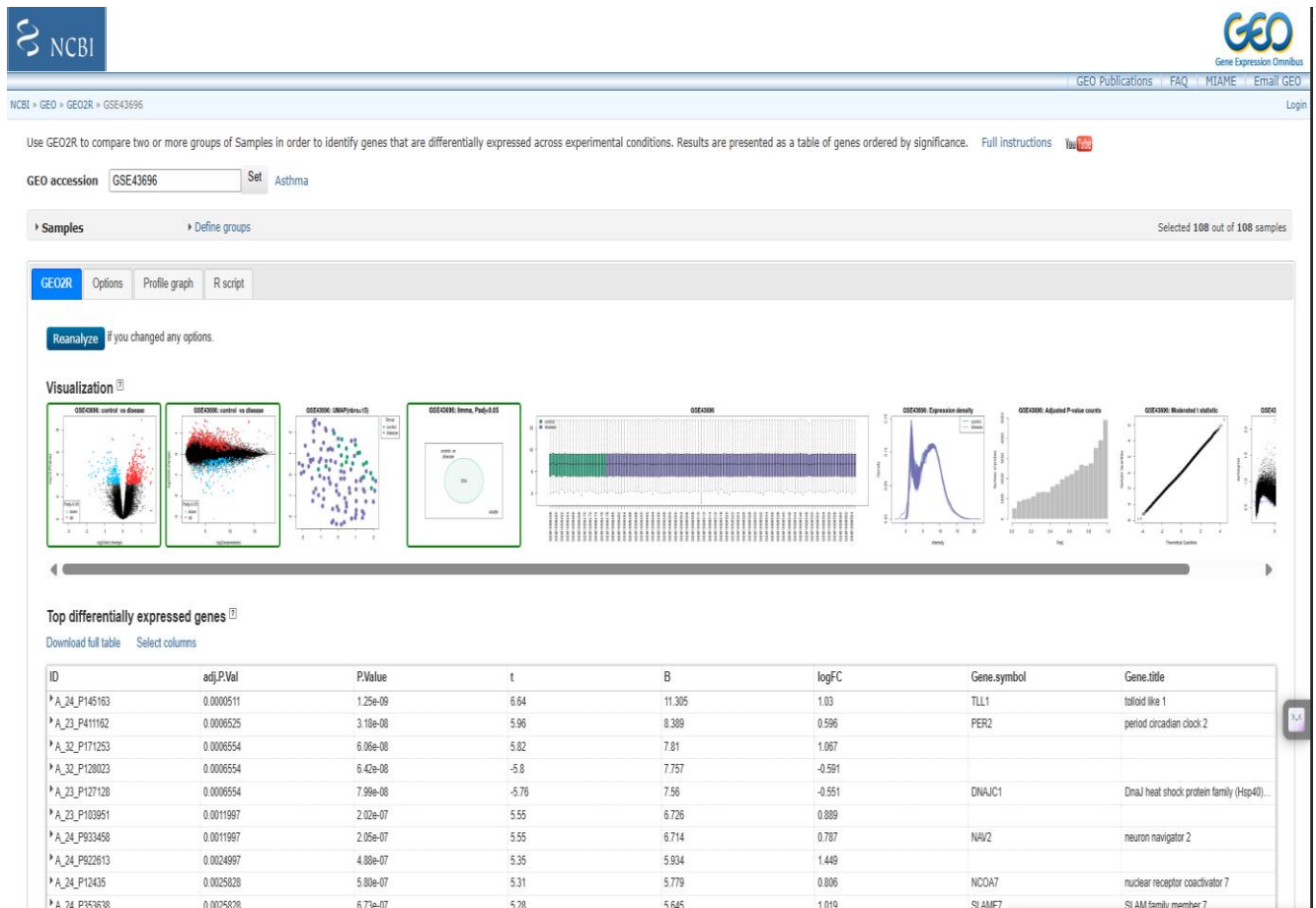
Status	Public on Jul 01, 2014
Title	Asthma
Organism	Homo sapiens
Experiment type	Expression profiling by array
Summary	Bronchial Epithelial Cells were isolated processed as described (Chu et al., 2002 and Zhao et al., 2011). The objective of the study was to identify differentially expressed genes between normal control (NC), mild-moderate asthmatic (MMA) and severe asthmatic (SA) patients.
Overall design	Isolated fresh bronchial epithelial cells were placed into trizol for further analysis.
Contributor(s)	Wenzel SE , Kaminski N , Milosevic J
Citation(s)	Voraphani N, Gladwin MT, Contreras AU, Kaminski N et al. An airway epithelial iNOS-DUOX2-thyroid peroxidase metabolome drives Th1/Th2 nitrative stress in human severe asthma. <i>Mucosal Immunol</i> 2014 Sep;7(5):1175-85. PMID: 24518246 Li X, Hawkins GA, Moore WC, Hastie AT et al. Expression of asthma susceptibility genes in bronchial epithelial cells and bronchial alveolar lavage in the Severe Asthma Research Program (SARP) cohort. <i>J Asthma</i> 2016 Oct;53(8):775-82. PMID: 27050946 Camiolo MJ, Zhou X, Wei Q, Trejo Bittar HE et al. Machine learning implicates the IL-18 signaling axis in severe asthma. <i>JCI Insight</i> 2021 Nov 8;6(21). PMID: 34591794

Analyze with GEO2R

Submission date	Jan 23, 2013
Last update date	Jan 07, 2022
Contact name	Jadranka Milosevic
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State/province	PA - Pennsylvania
ZIP/Postal code	15217
Country	USA

Platforms (1) [GPL6480](#) Affilent-014850 Whole Human Genome Microarray 4x44K G4112F

PIC 4) Differential Gene Expression Analysis (GEO2R) (GSE43696)



PIC 5)(A) Asthma Transcriptomics Presentation:

PHASE 6 · 2019–2024

Paper 1: Identifying Master Regulator Genes in Asthma

Network Study of Nasal Transcriptome Profiles Reveals Master Regulator Genes of Asthma

Published: August 2020 (JACI 2021)

Objective

Understand which genes control asthma rather than just identifying genes with altered expression.

Method

RNA-seq of nasal epithelial samples from children and adults with asthma. Used **WGCNA** and **Bayesian network analysis** to identify important gene networks.

Key Findings

FOXJ1 was identified as the major master regulator. Different asthma severities shared common gene-expression pathways. Gene networks related to cilia function and immune response were strongly altered.

Why it matters

This study shifted asthma research from studying **individual genes** to understanding **gene regulatory networks**.

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2019–2024

Phase 6 of 8

Do AN et al., J Allergy Clin Immunol. Published online: 20 Aug 2020; Print: 2021;147(3):879–893

PIC 5) (B) Asthma Transcriptomics Presentation:

PHASE 8 • 2024–2026 • NEW

Paper 3: Molecular Classification of Asthma

Multicohort Analysis of Bronchial Epithelial Cell Expression Reveals Four Clinically Distinct Clusters
Published: July 2025

Objective
To classify asthma patients using transcriptomic data.

Method
Combined data from **12 independent cohorts**
Bayesian multicohort meta-analysis
Gene-expression clustering

Key Findings
Identified a 505-gene asthma signature.
Asthma patients could be grouped into four molecular clusters.
Results were consistent across different populations.

Why it matters
Shows that asthma is not one disease but several molecular subtypes.

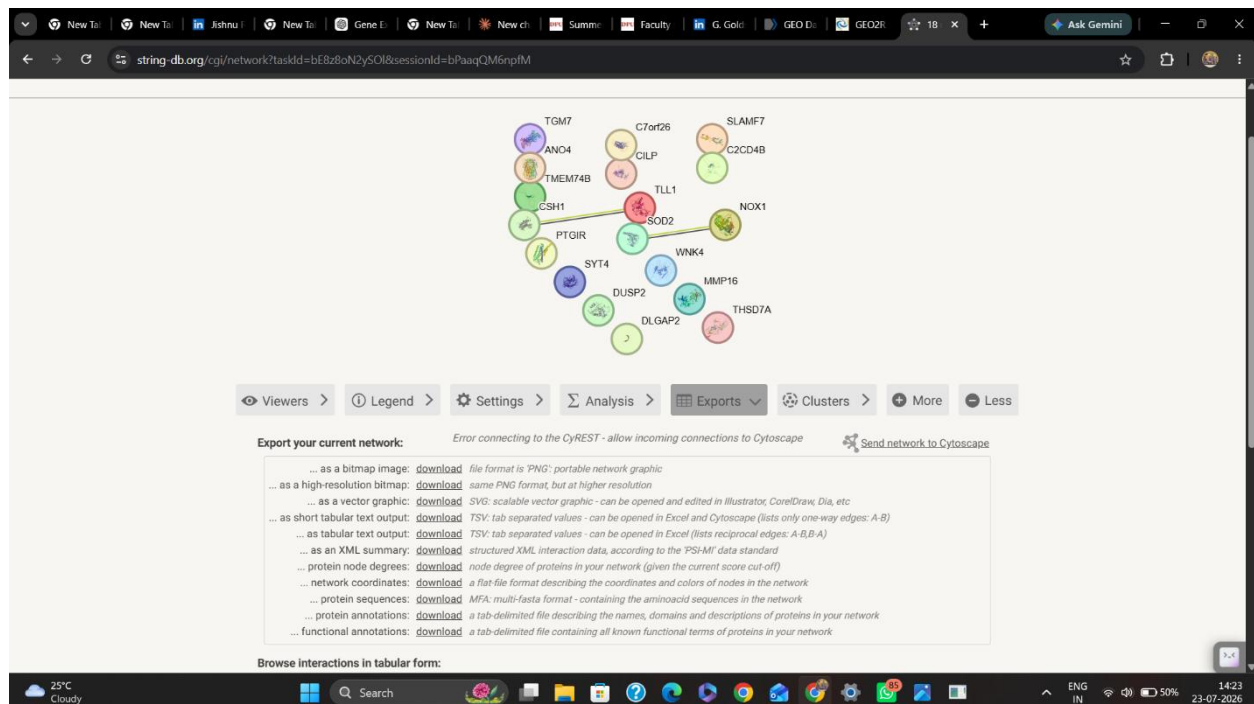


2024–2026
Phase 8 of 8
Lee I, Ganesan A, Kalesinskas L, et al., Am J Respir Cell Mol Biol. Jul 2025;73:73–87.

Evolution of Gene Expression Research in Asthma

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PIC 6) Protein–Protein Interaction Analysis STRING network:



string-db.org/cgi/network?taskId=bE8z8oN2ySOI&sessionId=bPaugQM6npfM

Export your current network:

- ... as a bitmap image: [download](#) file format is 'PNG': portable network graphic
- ... as a high-resolution bitmap: [download](#) same PNG format, but at higher resolution
- ... as a vector graphic: [download](#) SVG: scalable vector graphic - can be opened and edited in Illustrator, CorelDraw, Dia, etc
- ... as short tabular text output: [download](#) TSV: tab separated values - can be opened in Excel and Cytoscape (lists only one-way edges: A-B)
- ... as an XML summary: [download](#) TSV: tab separated values - can be opened in Excel (lists reciprocal edges: A-B, B-A)
- ... protein node degrees: [download](#) structured XML interaction data, according to the 'PSI-MI' data standard
- ... network coordinates: [download](#) node degree of proteins in your network (given the current score cut off)
- ... protein sequences: [download](#) a flat file format describing the coordinates and colors of nodes in the network
- ... protein annotations: [download](#) MFA: multi-fasta format - containing the aminoacid sequences in the network
- ... functional annotations: [download](#) a tab-delimited file describing the names, domains and descriptions of proteins in your network
- ... functional annotations: [download](#) a tab-delimited file containing all known functional terms of proteins in your network

Browse interactions in tabular form:

PIC 7) Functional Enrichment Analysis

GO results

AmiGO 2 Home Search Browse Tools & Resources Help Feedback About Quick search Search

Anti-Muellerian hormone

Gene Product Information

Symbol AMH
Name(s) Anti-Muellerian hormone
Type protein
Taxon Homo sapiens
Synonyms MIF
PAN-GO PAN-GO functions and evidence for Anti-Muellerian hormone
Database UniProtKB, P03971
Related [Link](#) to annotations page to AMH.
[Link](#) to annotations download for AMH.

Associations GO-CAMs (1)

Filter results

Total annotations: 40

No current user filters.

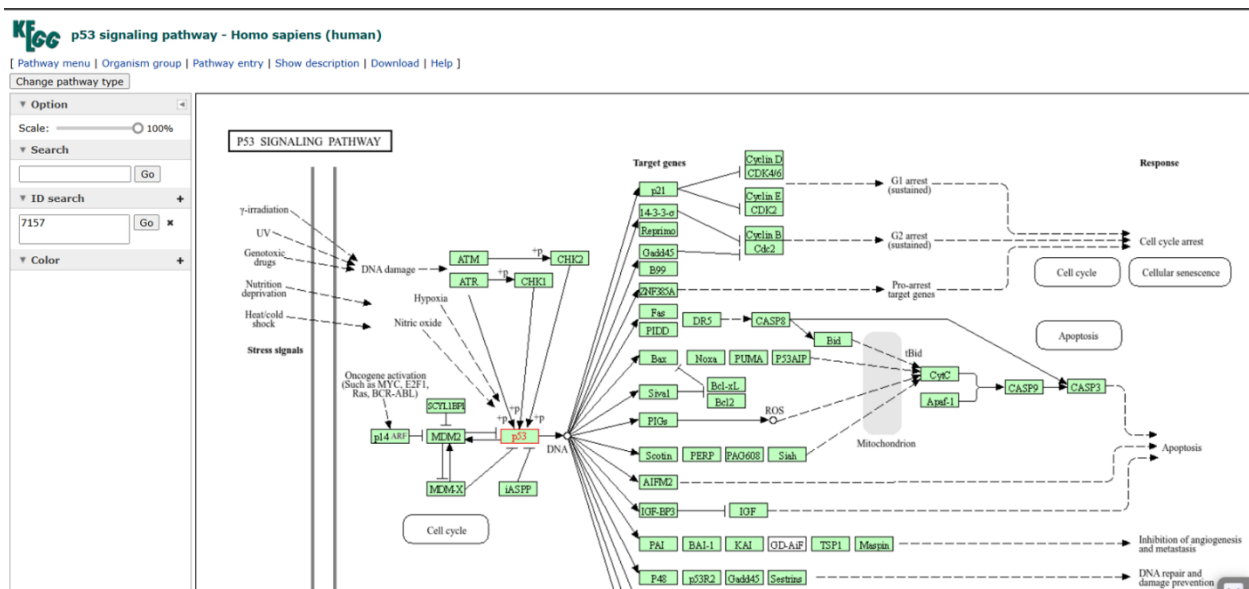
Total annotations: 40; showing: 1-10
Results count 10

«First» «Prev» «Next» «Last» Download (up to 100000)

Gene/product	name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Is of
☐	AMH	Anti-Muellerian hormone		transforming growth factor beta receptor	GO_Central	Homo sapiens	IBA	PANTHER:PTN002687475 RGD:2108 UniProtKB:P03971	muelierian-inhibiting factor	protein	UniProt

PIC 8) Functional Enrichment Analysis

KEGG pathway



PIC 9) Protein Annotation UniProt

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search Help

P69891 · HBG1_HUMAN

Function

Names & Taxonomy Protein: Hemoglobin subunit gamma-1 Amino acids: 147 (go to sequence)

Subcellular Location Gene: HBG1 Protein existence: Evidence at protein level

Disease & Variants Status: UniProtKB reviewed (Swiss-Prot) Annotation score: 55

PTM/Processing Organism: Homo sapiens (Human)

Expression Tools: Download Add Add a publication Entry feedback

Interaction Entry Variant viewer 147 Feature viewer Genomic coordinates Publications External links History

Structure

Family & Domains

Sequence

Similar Proteins

Homologs

Function

Gamma chains make up the fetal hemoglobin F₄ in combination with alpha chains. 3 Publications

Caution

The modification form of Leu-142 is subject of controversy and could be the artifactual result of sample handling. 1 Publication

Features

PIC 10) Protein Structure Analysis PDB

RCSB PDB Deposit Search Visualize Analyze Download Learn About Careers COVID-19 Help Contact us MyPDB

PDB-101 PDB EMDataResource NAKB wwPDB PDB-IMM f x t i n

Structure Summary Structure Annotations Experiment Sequence Genome Versions

Biological Assembly 1

1Y2A | pdb_00001y2a

Structure of mammalian importin bound to the non-classical PLSCR1-NLS

PDB DOI: <https://doi.org/10.2210/pdb1Y2A/pdb>

Classification: PROTEIN TRANSPORT

Organism(s): Mus musculus, Homo sapiens

Expression System: Escherichia coli BL21(DE3)

Mutation(s): No

Deposited: 2004-11-22 Released: 2005-02-01

Deposition Author(s): Chen, M.-H., Ben-Efraim, I., Mitrousis, G., Walker-Kopp, N., Sims, P.J., Cingolani, G.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.20 Å

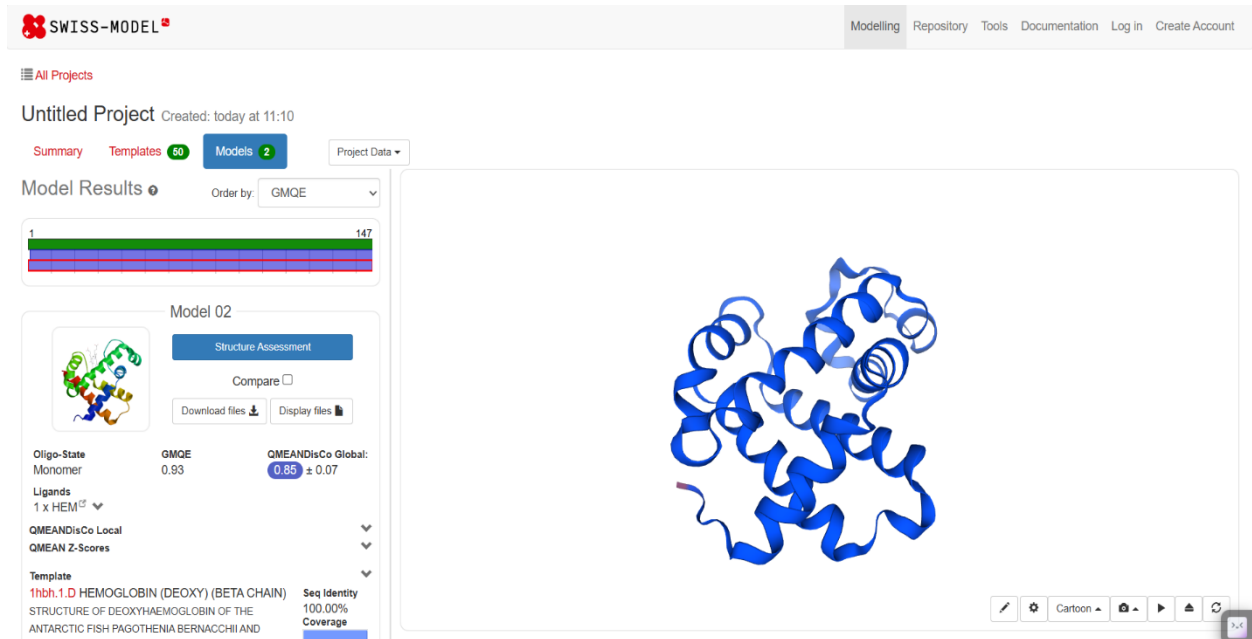
R-Value Free: 0.258 (Depositor), 0.269 (DCC)

wwPDB Validation

Metric	Percentile Ranks	Value
Rfree		0.269
Clashscore		20

PIC 11) Protein Structure Prediction

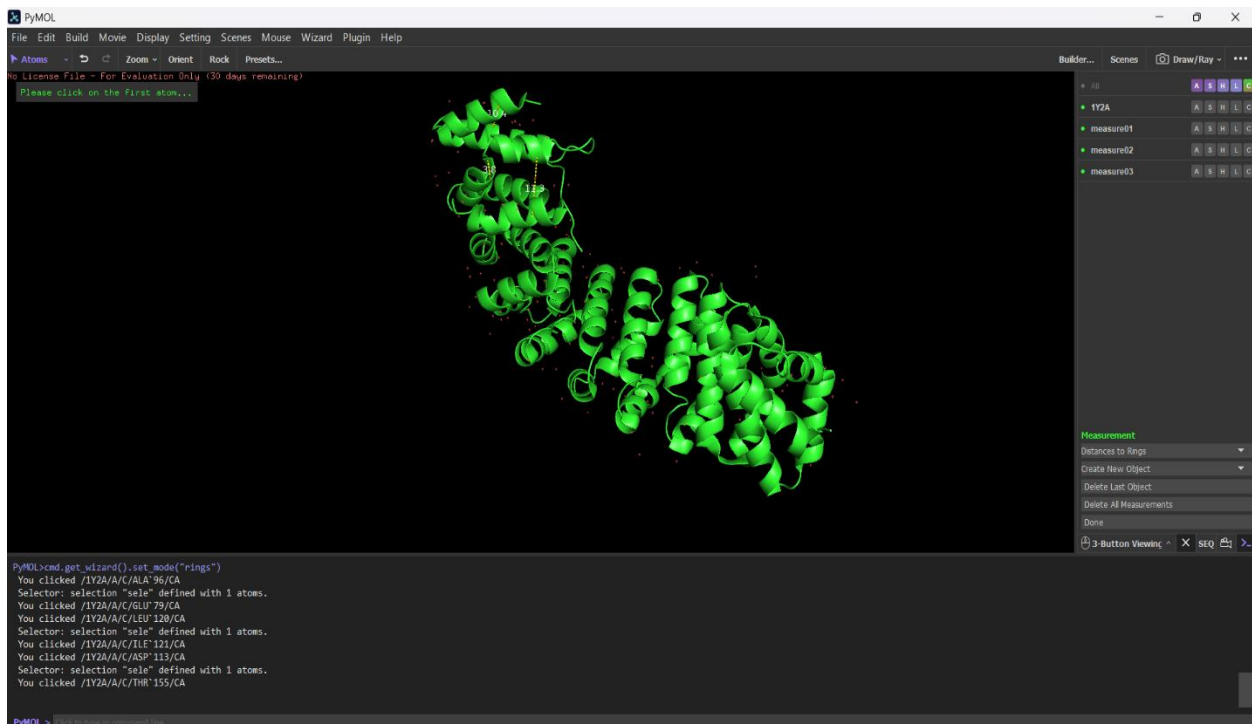
SWISS-MODEL:



The screenshot displays the SWISS-MODEL web interface. At the top, there's a navigation bar with links: Modelling, Repository, Tools, Documentation, Log in, and Create Account. Below this, the 'All Projects' section shows an 'Untitled Project' created today at 11:10. The project has 60 templates and 2 models. The 'Model Results' section shows a bar chart for the first model (1) and a detailed view for 'Model 02'. The detailed view includes a 'Structure Assessment' button, a 'Compare' checkbox, and 'Download files' and 'Display files' buttons. The 'Oligo-State' is 'Monomer', 'GMQE' is 0.93, and 'QMEANDisCo Global' is 0.85 ± 0.07. The 'Ligands' section shows '1 x HEM'. The 'QMEANDisCo Local' and 'QMEAN Z-Scores' sections are collapsed. The 'Template' section shows '1hbb.1.D HEMOGLOBIN (DEOXY) (BETA CHAIN)' with a 'Seq Identity' of 100.00% and 'Coverage' of 100.00%. The main 3D view shows a blue ribbon representation of the protein structure.

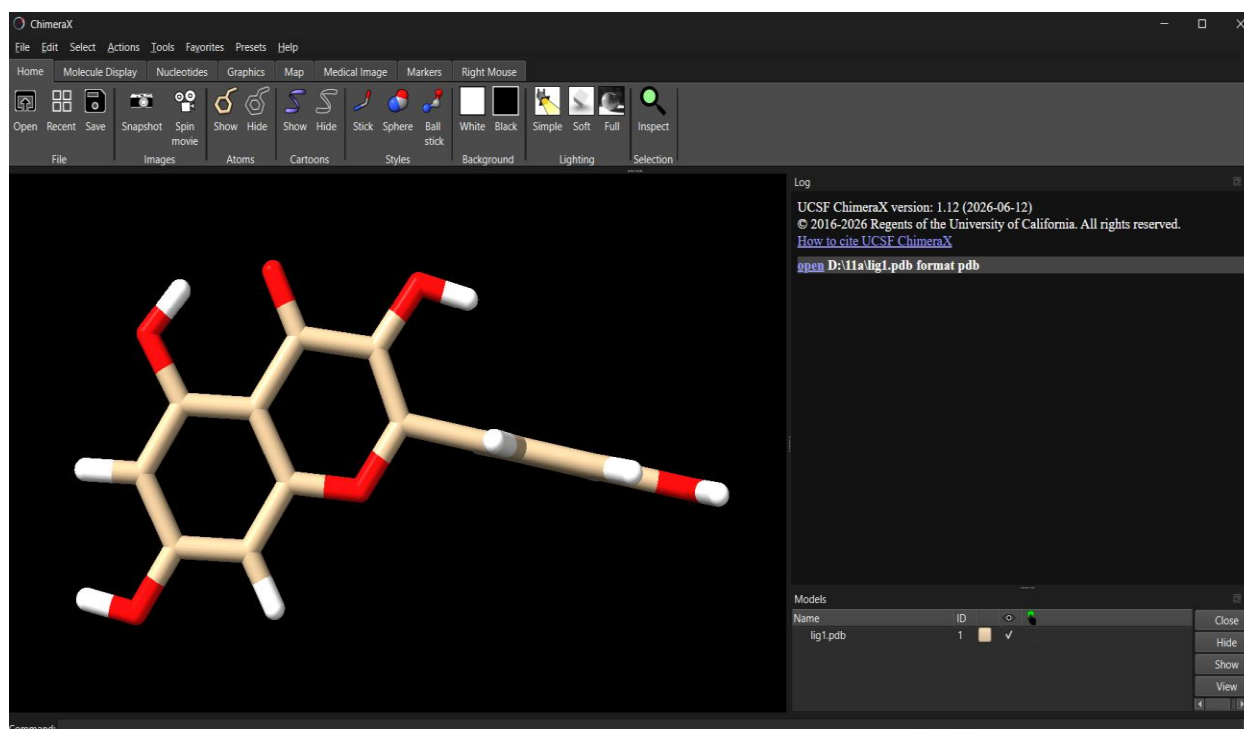
PIC 12) Protein Visualization

PyMOL

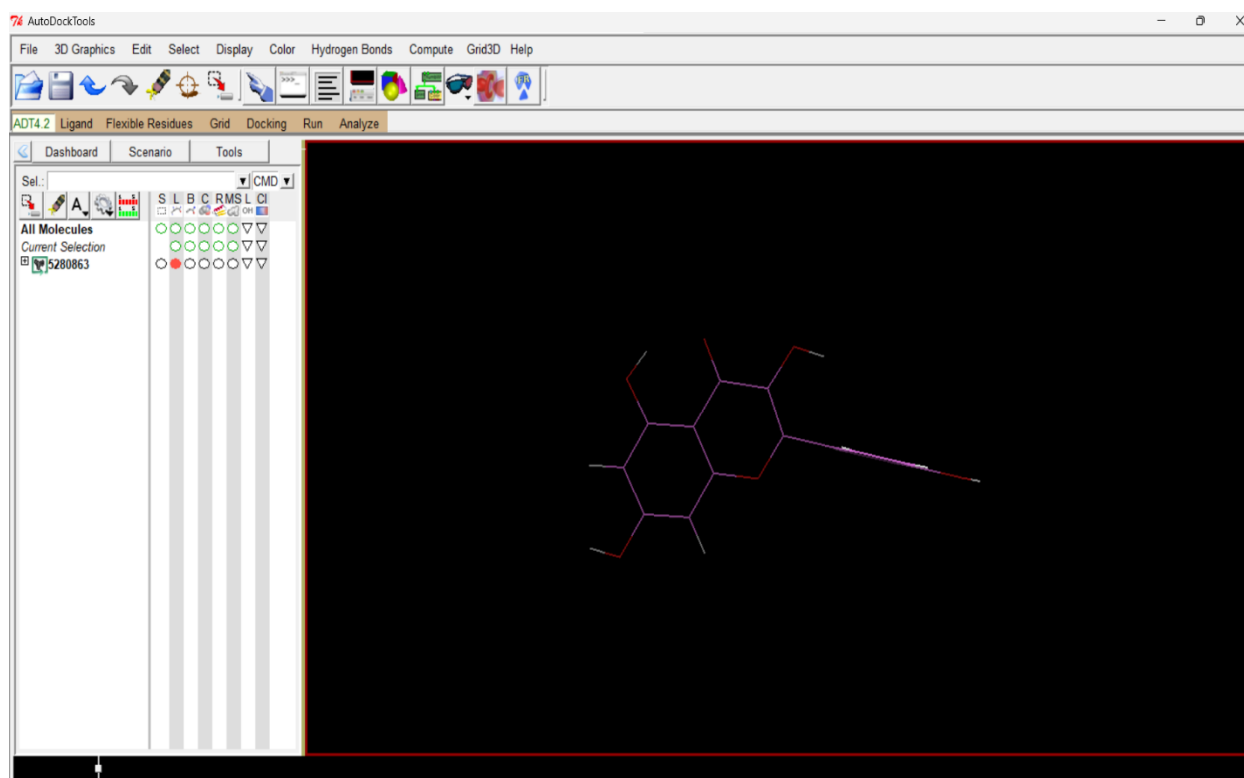


The screenshot shows the PyMOL software interface. The main window displays a green ribbon representation of a protein structure. The left sidebar shows the 'Atoms' panel with a list of atoms and a 'Please click on the first atom...' message. The right sidebar shows the 'Builder...' panel with a list of atoms and a 'Please click on the first atom...' message. The bottom panel shows the command line with the following commands: `PyMOL>cmd.get_wizard().set_mode("rings")`, `You clicked /1Y2A/A/C/ALA'96/CA`, `Selector: selection "sels" defined with 1 atoms.`, `You clicked /1Y2A/A/C/LEU'79/CA`, `You clicked /1Y2A/A/C/LEU'120/CA`, `Selector: selection "sels" defined with 1 atoms.`, `You clicked /1Y2A/A/C/ILE'121/CA`, `You clicked /1Y2A/A/C/ASP'113/CA`, `Selector: selection "sels" defined with 1 atoms.`, and `You clicked /1Y2A/A/C/THR'155/CA`. The bottom status bar shows 'PyMOL > Click to type or command line'.

PIC 13)(A) Molecular Docking Preparation ChimeraX



PIC 13)B)Molecular Docking Preparation AutoDock Tools



FUTURE PROSPECTS:

The knowledge and practical experience gained during this internship provide a strong foundation for advanced studies and research in bioinformatics, computational biology, structural biology, genomics, transcriptomics, and drug discovery. The rapidly growing integration of artificial intelligence, machine learning, and multi-omics technologies will continue to expand the role of bioinformatics in healthcare and biomedical sciences.

The computational approaches learned during the internship can be extended to cancer genomics, infectious diseases, precision medicine, pharmacogenomics, molecular diagnostics, vaccine development, and personalized therapeutics. Advanced applications such as genome editing, protein engineering, molecular dynamics simulations, systems biology, and AI-assisted drug discovery represent promising future research directions.

The experience gained in biological database exploration, gene expression analysis, protein structure prediction, molecular docking, and virtual screening will support future academic projects, higher education, research internships, and careers in biotechnology, pharmaceuticals, computational biology, and biomedical research. Furthermore, continued learning of programming, data science, and artificial intelligence will enhance the ability to develop more efficient computational methods for solving complex biological problems.

This internship has strengthened both technical knowledge and scientific reasoning, providing a solid platform for contributing to interdisciplinary research and future innovations in life sciences.

SUMMARY:

The Bioinformatics Summer Internship provided a comprehensive introduction to the integration of biology and computational science in modern biomedical research. Throughout the internship, I gained practical exposure to a wide range of bioinformatics databases, computational tools, and analytical techniques used in genomics, transcriptomics, structural biology, and computer-aided drug discovery.

The internship began with sessions on laboratory safety, scientific thinking, portfolio management, and research documentation, which established a strong foundation for the subsequent technical modules. I then explored several biological databases including NCBI, Gene Expression Omnibus (GEO), UniProt, Protein Data Bank (PDB), PDBsum, STRING, Gene Ontology (GO), KEGG, AlphaFold Protein Structure Database, and SWISS-MODEL to understand how biological information is stored, retrieved, and interpreted.

A major learning component involved transcriptomic analysis of asthma using publicly available GEO datasets. Differential gene expression analysis was performed using GEO2R, followed by functional enrichment analysis using GO, KEGG, and g:Profiler. Protein-Protein Interaction (PPI) networks were constructed using STRING and visualized using Cytoscape to identify hub genes and better understand disease-associated molecular pathways.

The internship also strengthened my understanding of structural bioinformatics through protein annotation, protein structure visualization, homology modelling, and structure prediction using UniProt, PDB, AlphaFold, SWISS-MODEL, and PyMOL. Practical sessions on molecular docking, virtual screening, protein preparation, ligand preparation, AutoDock Tools, AutoDock

Vina, and ChimeraX demonstrated how computational approaches are applied during the early stages of drug discovery and therapeutic target identification.

In addition to technical training, workshops on scientific writing, medicinal chemistry, stereochemistry, molecular interactions, and biostatistics enhanced my scientific communication skills and analytical thinking. These sessions emphasized the importance of interpreting computational results critically rather than relying solely on software-generated outputs. Overall, this internship significantly strengthened my understanding of computational biology by connecting gene expression analysis, functional annotation, protein structure analysis, network biology, and molecular docking into a single integrated workflow. The knowledge and practical experience gained during this internship have provided me with a strong foundation for future research in bioinformatics, computational biology, precision medicine, and structure-based drug discovery.

REFERENCES:

Biological Databases:

1. National Center for Biotechnology Information (NCBI). <https://www.ncbi.nlm.nih.gov/>
2. NCBI Gene Expression Omnibus (GEO). <https://www.ncbi.nlm.nih.gov/geo/>
3. GEO2R Analysis Tool. <https://www.ncbi.nlm.nih.gov/geo/geo2r/>
4. UniProt Protein Knowledgebase. <https://www.uniprot.org/>
5. Protein Data Bank (RCSB PDB). <https://www.rcsb.org/>
6. PDBsum. <https://www.ebi.ac.uk/thornton-srv/databases/pdbsum/>
7. STRING Database. <https://string-db.org/>
8. Gene Ontology Consortium. <https://geneontology.org/>
9. KEGG: Kyoto Encyclopedia of Genes and Genomes. <https://www.genome.jp/kegg/>
10. g:Profiler Functional Enrichment Tool. <https://biit.cs.ut.ee/gprofiler/>
11. AlphaFold Protein Structure Database. <https://alphafold.ebi.ac.uk/>
12. SWISS-MODEL Workspace. <https://swissmodel.expasy.org/>

Software:

13. Cytoscape (Version 3.10). <https://cytoscape.org/>
14. PyMOL Molecular Graphics System. <https://pymol.org/>
15. UCSF ChimeraX. <https://www.rbvi.ucsf.edu/chimerax/>
16. AutoDock Tools. <https://autodock.scripps.edu/>
17. AutoDock Vina. <https://vina.scripps.edu/>

Recommended Textbooks:

18. Lesk, A. M. *Introduction to Bioinformatics*. 5th Edition. Oxford University Press.
19. Mount, D. W. *Bioinformatics: Sequence and Genome Analysis*. 2nd Edition. Cold Spring Harbor Laboratory Press.
20. Pevsner, J. *Bioinformatics and Functional Genomics*. 3rd Edition. Wiley-Blackwell.
21. Lodish, H., Berk, A., Kaiser, C. A., et al. *Molecular Cell Biology*. W. H. Freeman.
22. Alberts, B., Johnson, A., Lewis, J., et al. *Molecular Biology of the Cell*. Garland Science.

Additional Learning Activities

Throughout the internship, several supplementary sessions and workshops were organized alongside the core bioinformatics training. These activities enhanced technical knowledge, improved professional skills, promoted workplace safety, and strengthened scientific communication. They provided valuable exposure to research methodologies, documentation practices, and analytical thinking that complement the practical aspects of bioinformatics.

Fire Safety and Emergency Drill

A fire safety and emergency preparedness session was conducted to familiarize students with essential safety practices within academic and research environments. The programme covered the basic causes of laboratory fires, preventive safety measures, emergency evacuation procedures, and the appropriate use of fire extinguishers.

Participants were instructed on recognizing different classes of fire, responding calmly during emergencies, identifying emergency exits, and following institutional safety protocols. A practical demonstration of fire extinguisher operation was also provided to improve awareness and preparedness. This session reinforced the importance of maintaining a safe working environment in laboratories and research facilities.

Mind Mapping and Portfolio Management Session

A dedicated session on mind mapping and portfolio management was organized to help students effectively document and present their internship activities. Students were introduced to various techniques for organizing ideas, connecting concepts, and planning scientific projects using structured mind maps.

The portfolio management component focused on preparing a professional internship report by organizing experimental work, screenshots, figures, tables, observations, and references in a systematic manner. Emphasis was placed on scientific writing, originality, logical organization, and maintaining proper documentation throughout the internship. The session helped students understand the importance of presenting technical work clearly and professionally.

Bioreactor Workshop

A hands-on **Bioreactor Workshop** was conducted to introduce students to the principles of upstream bioprocessing and large-scale cultivation of microorganisms and cells. The session provided practical exposure to the design, operation, and applications of laboratory-scale bioreactors used in biotechnology and biopharmaceutical industries.

Students were introduced to the major components of a bioreactor, including the **reaction vessel, impeller, agitator, sparger, baffles, sensors, probes, temperature control system, pH control system, dissolved oxygen (DO) monitoring system, and aeration mechanism**. The importance of maintaining sterile conditions and controlling environmental parameters during microbial growth was also explained.

The workshop demonstrated how critical process parameters such as **temperature, pH, dissolved oxygen, agitation speed, aeration rate, foam formation, and nutrient supply** are monitored and optimized to maximize cell growth and product formation. Students also learned

about the concepts of **batch, fed-batch, and continuous fermentation**, along with their industrial applications.

The session further highlighted the role of bioreactors in the production of **vaccines, antibiotics, enzymes, recombinant proteins, monoclonal antibodies, biofuels, and other biopharmaceutical products**. Practical demonstrations illustrated how biological processes can be scaled from laboratory experiments to industrial manufacturing while maintaining product quality and process efficiency.

Overall, the workshop provided valuable insight into the fundamentals of bioprocess engineering and demonstrated how bioreactors serve as a crucial bridge between laboratory research and large-scale industrial biotechnology production.

Scientific Writing Workshop

A scientific writing workshop was conducted to introduce students to the principles of writing clear, accurate, and well-structured scientific documents. The session covered the standard format of research papers, internship reports, review articles, and project documentation.

Students learned how to prepare effective titles, write concise abstracts, develop coherent introductions, describe methodologies accurately, present results logically, and draw meaningful conclusions. The workshop also emphasized the importance of plagiarism-free writing, proper citation practices, referencing scientific literature, and maintaining ethical standards in scientific communication. Practical tips for improving readability, grammar, and technical presentation were also discussed.

Statistics for Data Analysis Workshop

A workshop on biostatistics and data analysis was conducted to provide students with the statistical foundation required for analyzing biological datasets and interpreting research findings. The session introduced the importance of statistical methods in bioinformatics, biomedical research, and experimental biology.

Students learned descriptive statistical measures including **mean, median, mode, range, variance, standard deviation, quartiles, and interquartile range**, along with their applications in summarizing biological data. The concepts of data distribution, frequency analysis, graphical representation, and identification of outliers were also explained.

The workshop further covered **normal distribution, probability concepts, hypothesis testing, p-values, confidence intervals, correlation, and basic statistical significance**, enabling students to understand how statistical methods support scientific decision-making. Practical examples demonstrated how statistical analysis improves the reliability and interpretation of experimental results, particularly in gene expression studies, molecular biology experiments, and bioinformatics research.

Overall, the workshop strengthened students' ability to analyze biological data objectively, interpret quantitative results accurately, and apply appropriate statistical techniques in future research projects.

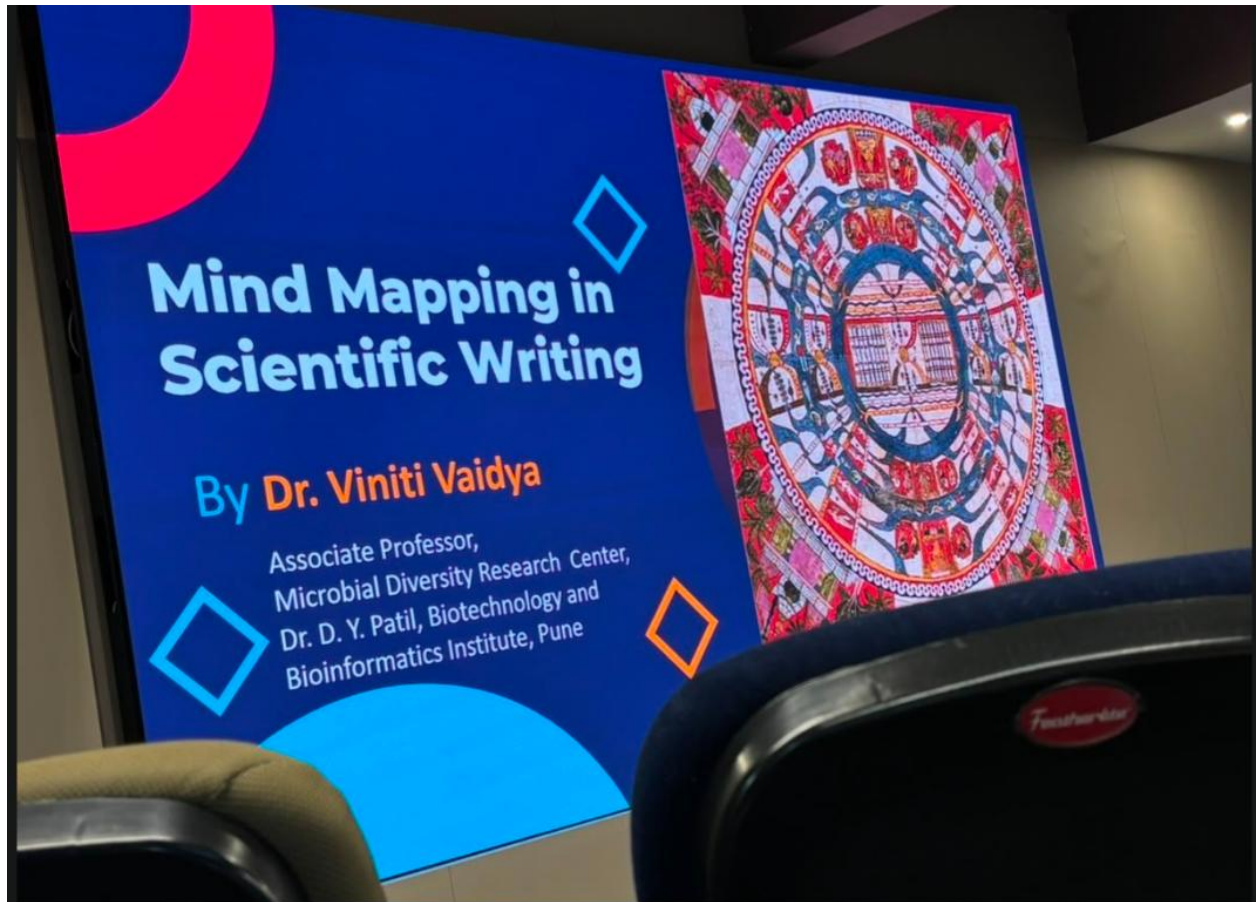
PIC 14) Picture of the Bioreactor used:



PIC 15) Portfolio Management Session:



PIC 16) Mind Mapping Session:



PIC 17) Laboratory Visit and Facility Exploration:

